

Analytical Pixel Integration of Gaussian Speckle Blobs

Renderer Convergence Conjecture

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1 Purpose

This note derives the pixel integral for the untruncated Gaussian blobs used in Experiment 2. It distinguishes the rigid and general affine cases, and records the consequences of the current radial cutoff and additive clamp.

2 Reference Speckle Field

Let $\mathbf{r} = (x, y)^T$ be reference coordinates, and let blob j have centre $\mathbf{c}_j$ and common standard deviation σ . Its untruncated coverage is

$$g_j(\mathbf{r}) = \exp\left(-\frac{\|\mathbf{r} - \mathbf{c}_j\|^2}{2\sigma^2}\right). \quad (1)$$

Ignoring overlap saturation for the moment, the total coverage and intensity are

$$G(\mathbf{r}) = \sum_j g_j(\mathbf{r}), \quad (2)$$

$$I(\mathbf{r}) = I_0 + \gamma [1 - 2G(\mathbf{r})]. \quad (3)$$

Thus a unit coverage is black, $I_0 - \gamma$, and zero coverage is white, $I_0 + \gamma$. The implemented renderer instead uses $G \leftarrow \min(G, 1)$; this is discussed in Section 7.

3 Pixel Mapping

Let $\mathbf{q} = (q_x, q_y)^T \in [-h, h]^2$ be coordinates relative to the centre of a target pixel, where $2h$ is the target-pixel width. On a rigid or affine patch, the inverse map from target to reference coordinates is

$$\mathbf{r}(\mathbf{q}) = \mathbf{r}_0 + \mathbf{A}\mathbf{q}. \quad (4)$$

Here $\mathbf{A}$ is the local inverse deformation gradient and $\mathbf{r}_0$ is the reference coordinate corresponding to the target pixel centre. The pixel average of blob j is

$$\bar{g}_j = \frac{1}{4h^2} \int_{-h}^h \int_{-h}^h g_j(\mathbf{r}_0 + \mathbf{A}\mathbf{q}) dq_y dq_x. \quad (5)$$

Define $\mathbf{d}_j = \mathbf{r}_0 - \mathbf{c}_j$. If $\mathbf{A}$ is nonsingular, define

$$\boldsymbol{\mu}_j = -\mathbf{A}^{-1}\mathbf{d}_j, \quad \mathbf{B} = \mathbf{A}^T \mathbf{A}, \quad \boldsymbol{\Sigma} = \sigma^2 \mathbf{B}^{-1}. \quad (6)$$

The exponent in Equation (5) becomes

$$g_j(\mathbf{r}(\mathbf{q})) = \exp\left[-\frac{1}{2}(\mathbf{q} - \boldsymbol{\mu}_j)^T \mathbf{B}(\mathbf{q} - \boldsymbol{\mu}_j)/\sigma^2\right]. \quad (7)$$

4 General Affine Result

Let $\Phi_2(\mathbf{z}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$ denote the bivariate normal CDF through upper limits $\mathbf{z}$. For the rectangle $R = [-h, h]^2$, let P_R be the usual inclusion–exclusion of its four CDF corners. Then

$$\int_R g_j(\mathbf{r}(\mathbf{q})) \, d\mathbf{q} = \frac{2\pi\sigma^2}{|\det \mathbf{A}|} P_R. \quad (8)$$

Equation (8) is exact for any nonsingular affine map. For a general shear, it requires a correlated bivariate normal CDF, rather than only the scalar error function.

5 Error-Function Form

If $\mathbf{B}$ is diagonal, write

$$\mathbf{B} = \text{diag}(b_x, b_y). \quad (9)$$

The target-coordinate Gaussian is then separable. Define

$$E_x = \text{erf}\left(\frac{\sqrt{b_x}(h - \mu_{j,x})}{\sqrt{2}\sigma}\right) - \text{erf}\left(\frac{\sqrt{b_x}(-h - \mu_{j,x})}{\sqrt{2}\sigma}\right), \quad (10)$$

$$E_y = \text{erf}\left(\frac{\sqrt{b_y}(h - \mu_{j,y})}{\sqrt{2}\sigma}\right) - \text{erf}\left(\frac{\sqrt{b_y}(-h - \mu_{j,y})}{\sqrt{2}\sigma}\right). \quad (11)$$

The exact blob integral is

$$\int_R g_j(\mathbf{r}(\mathbf{q})) \, d\mathbf{q} = \frac{\pi\sigma^2}{2\sqrt{b_x b_y}} E_x E_y. \quad (12)$$

For a rigid map, $\mathbf{A}$ is a rotation and therefore $\mathbf{B} = \mathbf{I}$, so Equation (12) applies directly. Translation-only rigid motion is the special case $\mathbf{A} = \mathbf{I}$. Axis-aligned affine scaling also remains separable. A general affine map with shear or non-orthogonal mapped axes does not: use Equation (8) instead.

6 Intensity Integral

If the blobs do not saturate when added, linearity gives

$$\bar{I} = I_0 + \gamma \left[1 - 2 \sum_j \bar{g}_j \right]. \quad (13)$$

Only nearby blobs need be considered in a numerical implementation. The untruncated expression has nonzero tails, but they decay exponentially and may be culled by a conservative tolerance without changing the formula.

7 Cutoff and Clamp Limitations

The present pattern truncates each blob radially at $k\sigma$:

$$g_j^{(k)}(\mathbf{r}) = g_j(\mathbf{r}) \mathbf{1}(\|\mathbf{r} - \mathbf{c}_j\| \leq k\sigma). \quad (14)$$

Under Equation (4), this adds an elliptical domain to the pixel rectangle. The integral is consequently over the intersection of a rectangle and an ellipse, and is not represented by the

simple products of error functions above. The fraction of a complete untruncated Gaussian’s integral discarded outside the radius is

$$\exp(-k^2/2) . \tag{15}$$

For $k = 4$, this is approximately 3.35×10^{-4} . Removing the hard cutoff is therefore the cleanest route to an analytic reference.

Likewise, the implemented $\min(\sum_j g_j, 1)$ is nonlinear. Where blobs overlap enough to saturate, the pixel must be partitioned by the saturation contour, so the linear sum of Equation (8) is no longer exact. The analytic path is exact when saturation is absent; otherwise a numerical correction or a different non-saturating composition rule is required.

8 Implementation Consequence

For a first analytic Gaussian renderer, use untruncated blobs and the linear sum. Use Equation (12) for rigid motion and diagonal affine maps. For a fully general affine deformation, evaluate the four terms of Equation (8) with a robust bivariate-normal CDF. Keep the current quadrature renderer as the reference for cutoff or clamp behaviour until their domain intersections are handled explicitly.